

ANIMAL PRODUCTION

- **Introduction to animal production**
 - Covers the importance of livestock in food security and economy
 - Animal behavior
 - Different animal production systems (Intensive, Extensive, and organic systems).
 - Overview of livestock types
 - Importance of livestock industry
 - Differences between indigenous and exotic breeds.
- **Animal Nutrition and Feeding**
 - Studies digestive systems (Digestive physiology of ruminants vs. Non – ruminants).
 - Nutrient requirements (carbohydrates, proteins, minerals, vitamins) for growth, reproduction, and production
 - Feed types (forages, concentrates and supplements)
 - Feed formulation and ration formulation techniques for different livestock species.
 - Fodder conservation (hay/silage)
- **Breeding and Genetic**
 - Introduction to Mendelian laws
 - Genetic principles: heredity, genotype, and phenotype
 - Selection methods and breeding systems (purebred, cross breeding).
 - Reproductive technologies: Artificial insemination (AI) and oestrus synchronization, and improving economic traits.
 - Record keeping and performance recording
- **Livestock Management Practices**
 - Routine management: Feeding, watering, handling, and grooming
 - Routine duties: Hoof trimming, docking, dehorning, castration, shearing, identification methods, and milking procedures.
- **Animal Health and Disease Control**
 - Preventive health care measures, vaccines, parasite control (dipping/spraying), and bio security measures
 - Zoonoses and “One Health” concept
 - Common diseases.

- **Production Systems (species – specific)**
 - **Housing and Environment:** Design and management of animal shelters, ventilation, hygiene, and environmental impact on production.
 - **Dairy Production:** Udder structure, milking procedures, and milk handling.
 - **Poultry Production:** Fowl/poultry breeds, broiler, and layer management.
 - **Beef and small Ruminants:** Sheep, goat, and camel management.
 - **Pig Production:** Housing, feeding, and breeding.
 - **Analysis of Production Systems:** Intensive vs. extensive systems, and organic animal production.
- **Agribusiness and Technology**
 - Economics and Marketing: Record keeping, business planning, farm finance,
 - Farm mechanization and structures.
 - Marketing and value addition of animal products.
 - Use of farm management software and computer applications.
 - Environmental impact and sustainability.

Key Skills Developed

- Gain practical skills in animal handling, vaccination, feed formulation, and farm business management.

Practical and Field Components

- Farm machinery maintenance and housing construction.
- Identification of farm machinery and tools
- Weight determination and livestock handling.
- Field attachment/internships (Industry experience)
- Practical feed formulation using local resources

Expected Outcomes

- This type of course is often structured into comprehensive units for certificates or diplomas in agriculture and animal science.
- Graduates will be prepared to manage livestock enterprises, improve animal productivity through scientific methods, and work in agricultural extension services, aiming for sustainable development and self-employment.